Waterfall Vs Agile – Comparison Table

Feature	Waterfall Model	Agile Model
Approach	Linear and sequential	Iterative and incremental
Process Flow	Moves step-by-step (Requirements → Design → Development → Testing → Deployment)	Continuous cycles (Plan → Develop → Test → Review → Repeat)
Flexibility	Rigid, changes are difficult once a phase is completed	Highly flexible; changes can be made anytime
Requirements	Must be clearly defined at the beginning	Can evolve throughout the project
Customer Involvement	Low; client reviews mainly at the end	High; client provides continuous feedback in every iteration
Delivery	Final product delivered after full development	Working increments delivered frequently (e.g., every Sprint)
Testing	Done after development is complete	Continuous testing within each iteration
Team Collaboration	Limited; roles are clearly defined and follow a structured hierarchy	High collaboration; cross-functional teams work together
Documentation	Extensive and mandatory at each stage	Minimal documentation; focuses on working software
Cost & Time	Higher cost if changes occur; timeline fixed	Cost-effective for changing requirements; timeline adaptive

Creating an Amazon Application Using the Waterfall Model

The **Waterfall Model** is a linear, sequential software development methodology.
Each phase must be completed **fully** before the next begins.
No going back.

The Waterfall phases (as shown in your image) are:

1. **Requirements Gathering**
2. **Design**
3. **Implementation**
4. **Testing**
5. **Deployment**
6. **Maintenance**

Below is the detailed step-by-step process for building an **Amazon online shopping application** using these phases.

REWORK: SCRUM & JIRA FRAMEWORK

Scrum

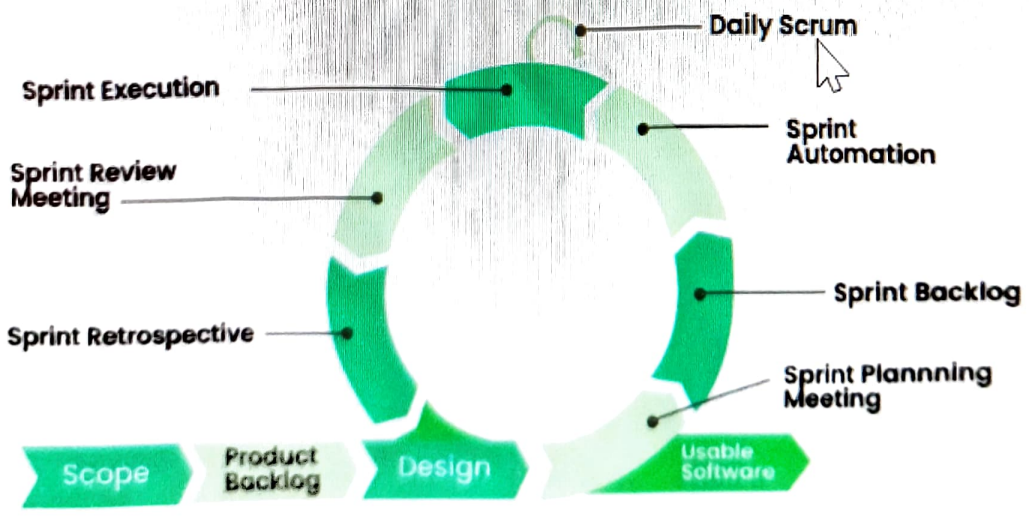
Scrum is an Agile framework used for developing complex products through collaboration, iterative progress, and continuous improvement. It organizes work into short, time-boxed cycles called *sprints* (usually 2–4 weeks), during which a cross-functional team delivers small, usable increments of the product. Scrum includes defined roles (Product Owner, Scrum Master, Development Team), key events (Sprint Planning, Daily Scrum, Sprint Review, Sprint Retrospective), and artifacts (Product Backlog, Sprint Backlog, Increment) that help teams stay focused, respond quickly to change, and continuously improve both the product and the process.

Three Pillars of Scrum

1. **Transparency:** Everyone should clearly see the work and process.
2. **Inspection:** The team regularly checks progress and the product.
3. **Adaptation:** The team adjusts and improves based on what they learn.

Scrum Roles

1. **Product Owner:**
   Represents the customer, decides what features are important, and manages the product backlog.
2. **Scrum Master:**
   Supports the team, removes obstacles, and ensures Scrum rules are followed.
3. **Development Team:**
   Builds and delivers a working piece of software at the end of each sprint.



Scrum Methodology

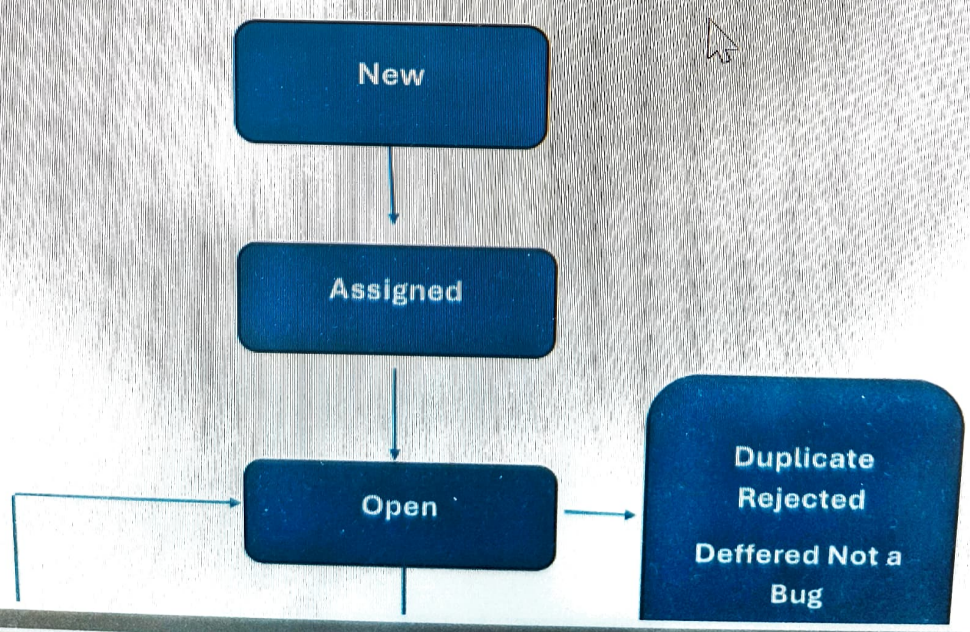
Scrum Events

1. **Sprint Planning:**

DEFECT / BUG LIFE CYCLE

- **A defect goes through different stages** from the moment it is found until it is fixed and closed.
- **Each stage shows the current status** of the defect so the team knows what is happening with it.
- **This cycle helps teams communicate clearly and work efficiently** to track, assign, fix, and verify bugs.
- **Defect status**
- **Defect status** shows the current stage a bug is in during the defect life cycle.
- It helps the team understand **how far the bug has progressed** toward being fixed.
- The main goal is to **track, communicate, and manage defects clearly and efficiently** throughout the project.

Defect States Workflow



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	A	B
	BRD No.	**BRD No.**
		User Account & Authentication
3	**BRD_01**	User lands on the homepage upon entering the valid application URL.
4	**BRD_02**	User can create an account using a valid email address.
5	**BRD_03**	System must validate email format and enforce password strength rules during sign-up.
6	**BRD_04**	User can log in with valid credentials (registered email + correct password).
7	**BRD_05**	User receives an error message for incorrect login credentials.
8		**Homepage Display**
9	**BRD_06**	Homepage must display the header, menu, search bar, and category list.
10	**BRD_07**	Homepage must show Login/Register options for guests.
11	**BRD_08**	Homepage must show My Account/Logout for logged-in users.
12	**BRD_09**	Homepage must display promotional banners or featured items (if configured).
13		**Registration**
14	**BRD_10**	User can navigate to the Registration page via header link.
15	**BRD_11**	Registration page must validate mandatory fields (First Name, Last Name, Email, Password).
16	**BRD_12**	System must validate duplicate email during registration.
17	**BRD_13**	Password and Confirm Password fields must match.
18	**BRD_14**	User must see a success message upon successful account creation.
19		**Login**
20	**BRD_15**	Login page must contain Email, Password fields and Login button.
21	**BRD_16**	System must validate email format before login attempt.